Madison Experimental Mathematics Lab: Cayley Graph Quotients and Twinless Paths

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Conjugacy and Almost Conjugacy

Definition (Conjugacy)

Two elements h and k of a group G are called *conjugate* if $ghg^{-1} = k$ for some $g \in G$. Similarly, two subgroups H and K of G are called *conjugate* if $gHg^{-1} = K$ for some $g \in G$.

Definition (Almost Conjugate Subgroups)

Two subgroups A and B of a finite group G are called *almost conjugate* if $|A \cap C| = |B \cap C|$ for every conjugacy class C in G.

Remark

Not every pair of almost conjugate subgroups is actually conjugate!

Well-Known Examples of Almost Conjugacy

Example 1

Consider the following subgroups A and B of the 3×3 special linear group $G = \mathrm{SL}_3(\mathbb{F}_2)$ with entries in the finite field $\mathbb{F}_2$:

$$A = \left\{ \begin{pmatrix} 1 & 0 & 0 \\ * & & \\ * & & \mathbf{A} \end{pmatrix} \in \mathrm{SL}_3(\mathbb{F}_2) : \mathbf{A} \in \mathrm{SL}_2(\mathbb{F}_2) \right\} \qquad B = A^{\mathrm{T}}$$

A and B are almost conjugate subgroups of G, but they are not conjugate.

Example 2

Consider the following subgroups A and B of the semi-direct product $G = (\mathbb{Z}/8\mathbb{Z})^{\times} \rtimes (\mathbb{Z}/8\mathbb{Z})$, whose multiplication behaves like $(a, b) \cdot (c, d) = (ac, b + ad)$:

$$A = \{(1,0), (3,0), (5,0), (7,0)\} \qquad B = \{(1,0), (3,4), (5,4), (7,0)\}$$

A and B are almost conjugate subgroups of G, but they are not conjugate.

Some New Examples of Almost Conjugacy

The Symmetric Group on Seven Letters

Inside the symmetric group S_7, the subgroups

$$A = \langle (1,2,7)(3,6,5), (2,4,7)(3,6,5) \rangle \qquad B = \langle (1,7)(4,5), (1,2,4)(5,7,6) \rangle$$

are almost conjugate but not actually conjugate.

Inside the Symmetric Group on Ten Letters

Inside the symmetric group S_{10}, the subgroups

$$A = \langle (3,6)(4,7), (1,2)(3,7)(4,6)(9,10), (1,10)(2,9)(3,6)(4,7) \rangle$$
$$B = \langle (1,7)(4,8)(5,9)(6,10), (1,10)(6,7), (1,7)(2,3)(4,8)(6,10) \rangle$$

are almost conjugate but not actually conjugate.

We have found even more examples in larger finite groups (almost 70 in total)!

The Code

Figure: QR Code to Github Repo

Cayley Graphs and their Quotients

Definition (Cayley Graph)

The *Cayley graph* $\mathrm{Cay}(G,T)$ of a group G with respect to a generating set T is the directed graph with vertex set G and edge set

$$\{(g,gt) \in G \times G : g \in G \text{ and } t \in T\} = \left\{(g,h) \in G \times G : g^{-1}h \in T\right\}.$$

That is, two vertices are connected by an edge if they differ by a generator.

Observation

A group G acts on its Cayley graph $\mathrm{Cay}(G,T)$ by left multiplication. The quotients

$$A\backslash\mathrm{Cay}(G,T) \qquad\qquad B\backslash\mathrm{Cay}(G,T)$$

by the actions of two almost conjugate (but not actually conjugate) finite index subgroups A and B of G are non-isomorphic edge-colored finite directed graphs.

Our Project

Observation (from previous slide)

A group G acts on its Cayley graph $\mathrm{Cay}(G,T)$ by left multiplication. The quotients

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by the actions of two almost conjugate (but not actually conjugate) finite index subgroups A and B of G are non-isomorphic edge-colored finite directed graphs.

Definition (Twinless Path)

A sequence of adjacent edges in a Cayley graph quotient is *twinless* if the edges are colored according to the pattern $a^m b^n$ for some generators $a, b \in T$ and integers $m, n \in \mathbb{Z}$.

Problem Statement

Given two almost conjugate (but not actually conjugate) subgroups A and B of a finite group G, can we distinguish the above Cayley graph quotients by counting *simple twinless paths*?

Thank you!

Questions?